MDPSA: Multi-Agent Dissolution–Precipitation Simulation Agent

Automated thermodynamic modeling of selective metal recovery

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NRCan CC-NRCan-078

Motivation

Problem

- Battery leachates (NMC532) contain Cu, Co, Ni as targets and Al, Fe, Mn as impurities.
- Selective precipitation via pH control requires a complexing modifier that retains target metals in solution while allowing impurities to precipitate.
- Identifying the best modifier $\log K$ combination is a high-dimensional search problem.

Solution

- Automate the full cycle: literature search $\rightarrow$ PHREEQC simulation $\rightarrow$ analysis $\rightarrow$ optimization.
- Use a LangGraph multi-agent pipeline so each stage is modular, reproducible, and auditable.
- Close the loop with Bayesian optimization to suggest improved $\log K$ values without exhaustive enumeration.

Experimental reference: NRCan CC-NRCan-078. Leachate composition: Cu, Co, Ni, Mn, Al, Fe in H_2SO_4 medium. Modifier dose 0.20 mol/kgw.

Pipeline Architecture

Agent	Role
ScoutAgent	Queries Claude LLM for NIST formation constants; writes `.phr` (PHREEQC reaction block) and `.yaml` (species tracking list).
SimRunnerAgent	Runs PHREEQC titration ($0 \rightarrow 1.0$ equiv NaOH, 60 steps) for open, closed, and partial CO_2 scenarios; parses output CSVs.
AnalystAgent	Computes dissolved-metal fractions, RMSE vs. experiment, selectivity scorecard, and species-contribution figures.
BOAgent	Gaussian-process Bayesian optimization over $\log K$ space; proposes next modifier candidate.
BeamerAgent	Assembles and compiles this XeLaTeX PDF automatically at end of each run.

Modifier files in `database/modifiers/`; simulation outputs in `experiments/single/`; figures and PDFs in `presentations/{modifier}/`.

Orchestration

- **Framework**: LangGraph `StateGraph`
- **State**: typed `SimState` dict passed between nodes
- **Routing**: conditional edges; SimRunner retries on PHREEQC error (max 3)
- **LLM**: Anthropic Claude (claude-sonnet-4-5) via Messages API

Flow

Scout $\rightarrow$ SimRunner $\rightarrow$ Analyst $\rightarrow$ BO $\rightarrow$ Beamer

Geochemical Model

PHREEQC setup

- Thermodynamic database: `sit.dat` (SIT activity model)
- NaOH titration: $0 \rightarrow 1.0$ equiv in 60 kinetic steps
- Equilibrium phases allowed to precipitate: Gibbsite, Ferrihydrite(am), $Cu(OH)_2(s)$, $Co(OH)_2(s)$, $Ni(OH)_2(s)$, MnOOH
- Three CO_2 scenarios: open (atmosphere), closed (sealed), partial (0.1 atm)
- Fe complexation suppressed (kinetic limitation)

Complex types in `.phr`

Type	Reaction	Example
ML	$M^{n+} + L = ML$	$CuEdta^{2-}$
ML_2	$M^{n+} + 2L = ML_2$	$CuEdta_2^{6-}$
MOHL	$M + L - H^+ + H_2O =$ product	CuOHEdta

- $\log K$ values from NIST SRD 46 v8.0
- Values tuned iteratively against experiment
- BO closes the tuning loop automatically

NIST Standard Reference Database 46 Version 8.0. SIT: Specific Ion Interaction Theory. Temperature: 25°C.

Selectivity Scoring and Optimization

Scorecard logic

- Comparison points: 0.3 and 1.0 equiv NaOH
- **Target metals** (Cu, Co, Ni): pass if dissolved fraction $> 70\%$ at each comparison point
- **Impurity metals** (Al, Fe, Mn): pass if dissolved fraction $< 30\%$
- Score: 0, 1, or 2 per metal (one point per equiv point)
- RMSE computed over all experimental equiv points

Bayesian optimization

- Objective: maximize total selectivity score
- Search space: $\log K$ of each complex ± 3 units around NIST reference
- Surrogate: Gaussian process with Matérn kernel
- Acquisition: expected improvement (EI)
- Each pipeline run adds one observation; BO proposes the next $\log K$ set

BO implemented with `scikit-learn` GaussianProcessRegressor. Modifier dose and leachate composition fixed; only $\log K$ values are optimized.

Outputs and Reproducibility

Per-run outputs

File	Content
`.phr`	PHREEQC reaction block
`.yaml`	Species tracking list
`neutralization.csv`	Titration results
`*_model_vs_exp.png`	Model vs. experiment
`*_species_contribution.png`	Species breakdown
`*_summary.pdf`	This presentation
`run_config.yaml`	Full run history

Reproducibility

- All modifier files version-controlled in Git
- `run_config.yaml` logs every run with scorecard, figure paths, and timestamp
- `--force-scout` flag regenerates LLM output; omitting it reuses existing `.phr`/`.yaml`
- Presentations committed to `presentations/{modifier}/`
- `.tex` files Overleaf-ready (`% !TEX program = xelatex`)

Repository: `github.com/vahid2364/modifier-driven-precipitation-simulation-agent`. Stack: Python 3.11, LangGraph, PHREEQC v3, scikit-learn, XeLaTeX (TeX Live 2026).